Configuration Guide

Low-Power Operation, UART Baud Rate,
and Application Sampling Interval

MSP430FR6043 Ultrasonic Software Library

Repository path: `workspace_ccstheia/Test_Code`

Document type: Engineering technical reference.
Audience: Firmware developers maintaining USS application timing, power profile, and debug telemetry.
Typesetting: pdfLaTeX with `newtx` (Times-compatible).

April 11, 2026

Executive summary

This guide specifies how to *enable* or *disable* application-level low-power behavior, how to reconfigure the eUSCI UART bit rate, and how to set the periodic measurement cadence (often called "sampling rate" in this project). Each item is tied to concrete files and symbols under `Test_Code`. Texas Instruments USS library internals may still use brief low-power intervals during capture or fault handling; this document addresses only the paths under direct application control.

Contents

Executive summary **1**

1 Scope and definitions **2**
- 1.1 Application-level low power . . . 2
- 1.2 Sampling cadence (output data rate) . . . 2
- 1.3 Out of scope for "disabling LPM" . . . 2

2 Enabling low-power operation **2**
- 2.1 Main loop (`main.c`) . . . 2
- 2.2 Offset calibration (`main.h`, `main.c`) . . . 2
- 2.3 Summary — low-power profile . . . 3

3 Disabling application low-power operation **3**
- 3.1 Main loop (`main.c`) . . . 3
- 3.2 Offset calibration . . . 3
- 3.3 Library behavior . . . 3
- 3.4 Summary — active profile . . . 3

4 UART baud rate reconfiguration **4**
- 4.1 Procedure . . . 4

5 Application sampling cadence **4**
- 5.1 Primary symbol . . . 4
- 5.2 Timer binding . . . 4
- 5.3 Documentation hygiene . . . 4
- 5.4 Serial throughput . . . 4

6 Quick reference **5**

References **5**

1 Scope and definitions

1.1 Application-level low power

Three independent mechanisms govern what the *application* does between samples:

1. **Capture API:** `USS_startLowPowerUltrasonicCapture` (low-power USS sequence) versus `USS_startUltrasonicMeasurement(..., USS_capture_power_mode_active)` (active CPU during acquisition).
2. **Inter-measurement wait:** second argument to `USS_waitForAppTimerElapse`, selecting either a low-power mode (e.g. LPM3) or `USS_low_power_mode_option_active_mode`.
3. **Optional offset calibration** (when enabled): member `isUseLPMCapture` in `USS_dTof_absTof_offset_te` controlled by `APPLICATION_ABSTOF_DTOF_OFFSET_LPM_CAPTURE` in `main.h`.

1.2 Sampling cadence (output data rate)

The main-loop period is derived from the USS system measurement interval, expressed in **LFXT cycles** as `USS_SYS_MEASUREMENT_PERIOD` in `USS_Config/USS_userConfig.h`. After configuration, `USS_configAppTimerPeriod` loads this value so successive iterations are spaced by approximately

$$T_s = \frac{\texttt{USS_SYS_MEASUREMENT_PERIOD}}{\texttt{USS_LFXT_FREQ_IN_HZ}}\ \text{s}, \qquad f_s \approx \frac{1}{T_s}. \tag{1}$$

For the common case `USS_LFXT_FREQ_IN_HZ` = 32768,

$$\texttt{USS_SYS_MEASUREMENT_PERIOD} \approx \text{round}\left(\frac{32768}{f_s}\right) \quad \text{with } f_s \text{ in Hz.} \tag{2}$$

The library documents an allowable range for this parameter (on the order of 20 ms to 2 s); remain within that range when changing rates.

1.3 Out of scope for "disabling LPM"

- `PM5CTL0 &= ~LOCKLPM5` in `system\_pre\_init.c` releases GPIO lock after power-up on FR devices. It is **not** a CPU sleep mode. Do **not** remove it when switching to an active power profile.
- SDHS acquisition rate, digital filtering, and USS pulse configuration in `USS\_userConfig.h` are distinct from the application cadence f_s treated here.

2 Enabling low-power operation

To reduce energy between measurements and follow the USS low-power capture flow:

2.1 Main loop (`main.c`)

- Invoke `USS_startLowPowerUltrasonicCapture(&gUssSWConfig);`
- After optional UART transmission, call
  `USS_waitForAppTimerElapse(&gUssSWConfig, USS_low_power_mode_option_low_power_mode_3);`
  or `USS_low_power_mode_option_low_power_mode_0/1/2` per system constraints (see TI `ussSwLib.h`).

2.2 Offset calibration (`main.h`, `main.c`)

When `APPLICATION_ENABLE_ABSTOF_DTOF_OFFSET_CALIBRATION` is `true`:

- Set `APPLICATION_ABSTOF_DTOF_OFFSET_LPM_CAPTURE` to `true`.

- Keep `.isUseLPMCapture = APPLICATION_ABSTOF_DTOF_OFFSET_LPM_CAPTURE` in the `abstoFDtofTestConfig` initializer.

2.3 Summary — low-power profile

Objective	Artifact	Setting
Low-power USS capture	`main.c`	`USS_startLowPowerUltrasonicCapture`
Sleep while waiting for tick	`main.c`	`USS_waitForAppTimerElapse(..., USS_low_power_mode_option_low_power_mode_3)`
LPM-friendly offset routine	`main.h` / `main.c`	`APPLICATION_ABSTOF_DTOF_OFFSET_LPM_CAPTURE == true`

3 Disabling application low-power operation

To keep the MCU in an application-controlled active profile between samples:

3.1 Main loop (`main.c`)

- Use `USS_startUltrasonicMeasurement(&gUssSWConfig, USS_capture_power_mode_active);`
- Use `USS_waitForAppTimerElapse(&gUssSWConfig, USS_low_power_mode_option_low_power_mode_0);` (shallowest LPM the API supports while waiting; there is no `active_mode` enumerator in this library)

3.2 Offset calibration

If calibration is compiled in, set `APPLICATION_ABSTOF_DTOF_OFFSET_LPM_CAPTURE` to `false`, or assign `.isUseLPMCapture = false` explicitly.

3.3 Library behavior

The TI USS sources may still use short LPM segments internally. The combination above is the standard **application active** pattern; removing all LPM from the library is neither required nor recommended.

3.4 Summary — active profile

Objective	Artifact	Setting
Active USS capture	`main.c`	`USS_startUltrasonicMeasurement(..., USS_capture_power_mode_active)`
Lightest timer wait (LPM0)	`main.c`	`USS_waitForAppTimerElapse(..., USS_low_power_mode_option_low_power_mode_0)`
Non-LPM offset capture	`main.h` / `main.c`	`APPLICATION_ABSTOF_DTOF_OFFSET_LPM_CAPTURE == false`

4 UART baud rate reconfiguration

The eUSCI block is programmed in `system\_pre\_init.c` when `APPLICATION_ENABLE_UART_DEBUG` is non-zero. The achieved baud rate is a function of BRCLK (here, **SMCLK at 8 MHz**) and the divider/-modulator fields `UCA1BRW`, `UCA1MCTLW`.

4.1 Procedure

1. **Record the target rate** in `main.h`:

```
#define APPLICATION_UART_BAUDRATE   115200
```

2. **Program matching hardware values** in `system\_pre\_init.c`. Examples for 8 MHz SMCLK:

```
/* 115200 baud */
UCA1BRW = 4;
UCA1MCTLW = 0x5551;

/* 230400 baud */
UCA1BRW = 34;
UCA1MCTLW = (0x54 << 8) | (11 << 4) | UCOS16;
```

3. **Validate BRCLK.** If SMCLK is no longer 8 MHz, recompute `UCA1BRW` / `UCA1MCTLW` using the device User's Guide or TI's baud calculator.
4. **Host tool:** configure the PC terminal for the same baud rate, 8N1, matching `UCA1`.

5 Application sampling cadence

5.1 Primary symbol

Edit `USS\_Config/USS\_userConfig.h`:

```
#define USS_LFXT_FREQ_IN_HZ          32768   /* must match crystal */
#define USS_SYS_MEASUREMENT_PERIOD   164   /* e.g. ~200 Hz at 32.768 kHz */
```

5.2 Timer binding

`main.c` calls:

```
USS_configAppTimerPeriod(&gUssSWConfig,
                         gUssSWConfig.systemConfig->measurementPeriod);
```

The structure field is populated from `USS_SYS_MEASUREMENT_PERIOD` via `USS\_userConfig.c`; rebuild after any change to the macro.

5.3 Documentation hygiene

Align comments and `APPLICATION_OUTPUT_DATA_RATE_HZ` in `main.h` and the banner in `main.c` with the realized f_s.

5.4 Serial throughput

With debug UART enabled, verify that (bytes transmitted per iteration) $\times f_s$ stays within the configured UART capacity including framing overhead.

6 Quick reference

Task	Location
Enable application LPM	`main.c`: low-power capture + LPM argument to `USS_waitForAppTimerElapse`
Disable application LPM	`main.c`: `USS_startUltrasonicMeasurement(..., active)` + `USS_waitForAppTimerElapse(..., LPM0)`
Change baud	`main.h`: `APPLICATION_UART_BAUDRATE`; `system\_pre\_init.c`: `UCA1BRW`, `UCA1MCTLW`
Change cadence f_s	`USS\_userConfig.h`: `USS_SYS_MEASUREMENT_PERIOD`; refresh comments in `main.h`, `main.c`

References

- Texas Instruments, *MSP430FR604x Family User's Guide* — UART baud-rate generation.
- Texas Instruments, *USS Software Library User's Guide* — `USS_startLowPowerUltrasonicCapture`, `USS_waitForAppTimerElapse`, system measurement period.